



OPEN New design of plate implant for bone healing assessment using frequency measurement: a biomechanical cadaveric study

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This study is a biomechanical cadaveric investigation to assess bone healing by measuring frequency changes using a new plate design called bone union detection with wire's natural frequency (BUDWF). Based on natural frequency and strain theory, in bone fracture conditions, the plate experiences higher strain under axial load and generates higher frequency, compared to the bone union condition where strain is lower and frequency is lower. If frequency changes can be detected, they can be used to determine the bone healing status. Eight human cadaveric tibia bones were used in the experiment. Bone union and fracture with a 10 mm gap, fixed by the BUDWF plate, were tested under various axial loads (0 N, 100 N, 200 N, 300 N, and 400 N) using the "Instron Electropuls E10000". We found that the frequency did not significantly change with any axial loads in bone healing, which is different from the bone fracture where the frequency was higher under high axial load. When comparing the two bone conditions, there were significant differences in frequency change under axial loads of 200, 300, and 400 N ($p < 0.001$). From the result, The BUDWF plate with frequency measurement can assess bone healing by comparing the frequency changes from the bone fracture and the bone union condition under an axial load of 400 N. We hope this plate will be the prototype for developing the device to help bone healing detection in the future.

Keywords Bone healing, Union, Plate, Acoustic, Frequency, Strain

Bone fractures are common injuries in orthopedics. In 2019, the global incidence of new fracture cases was approximately 178 million, with most fractures occurring in the lower leg and ankle¹. In the United States, the incidence of fractures was 4,017 cases per 100,000 population per year from 2009 to 2011, reflecting an 11% increase compared to the 3,627 cases per 100,000 population per year recorded during previous years². In Sweden, the incidence of tibial fractures was 51.7 per 100,000 population per year³. The primary goal of treatment is to achieve complete bone union. However, complications such as nonunion are common and lead to quality-of-life and financial issues. The incidence of nonunion ranges from 1.9% to 10%^{4–9}. Previous studies indicated that the overall cost of treating fracture healing complications was AUD 4.9 million¹⁰ and approximately CN\$11,800 per case, including direct and indirect costs¹¹.

Detecting bone union is important and involves various methods, including physical examination, plain radiography, ultrasound, CT scan, MRI, and bone scans¹². Plain radiography is the most commonly used method for detecting bone union and includes scoring systems such as RUST¹³, modified RUST¹⁴, and RUSH¹⁵. However, detecting bone union remains challenging in some cases, and misdiagnosing bone union can lead to complications such as bone nonunion and implant failure. From this reason, many previous studies have shown the development of methods for detecting bone union, such as strain gauges^{16,17}, displacement of tungsten¹⁸,

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fluid level¹⁹ or contrast media displacement²⁰, PZT voltage²¹, wavelengths²², resonance response frequency (RRF)²³, Force sensor²⁴, and the latest advancement involves detecting changes in wire natural frequency²⁵.

The concept of bone union detection using a wire's natural frequency (BUDWF) plate was to detect changes in wire natural frequency as a means of assessing strain changes in the plate. When the strain on the plate was high, the frequency was also high (high tension of the wire). Conversely, the frequency was also low (low tension of wire) when the strain on the plate was low. In a bone fracture, the strain on the plate increased when axial load was applied to the bone; simultaneously, the strain on the plate decreased as the bone gradually healed. Therefore, if the frequency decreases, it indicated that the strain was also decreasing, which confirmed that bone healing was improving. The concept was as shown in Fig. 1.

We hypothesize that changing the wire's natural frequency on the bone union detection using a wire's natural frequency (BUDWF) plate²⁵ can assess bone healing.

Materials and methods

Plate implant design

The bone union detection using a wire's natural frequency (BUDWF) plate²⁵ was modified from a 10-hole narrow DCP plate (DePuy-Synthes) (dimension of 167 mm × 12 mm × 4 mm). The middle four holes were connected to create a slot (7 × 60 mm) for applying a wire with free movement. The 316 L stainless steel wire, diameter 0.3 mm, was laser welded to the wire connector at the proximal and distal borders of the space in the longitudinal direction, centered within the space. The frequency of the wire after laser welding was set between 2000 and 3500 Hz, which was above the environmental noise frequency (500–2000 Hz) and proper for the length and diameter of the wire. The sound generation was made from polyetheretherketone (PEEK), which had a good elastic modulus and biocompatibility. The bottom middle of the sound generator served as the pick edge for strumming the wire when pressing down on the top of the device. The sound generator was attached to the BUDWF plate with two screws. The sound was created by manually pressing down on the center of the sound generator at the pick edge level to strum the wire by the pick edge. A picture of the BUDWF plate is shown in Fig. 2.

Concept of bone union detection with BUDWF plate²⁵

The wire's natural frequency corresponds to the tension in the wire and the load on the fixation plate. When axial load was applied to the bone and plate, tension occurred in the wire of the plate. If the bone was fractured, the load and strain on the plate were high, creating high tension in the wire, which produced a high-frequency sound when detected with a sound generator. In contrast, when the bone healed, the load and strain on the bone were low, resulting in lower tension in the wire and a lower frequency. Thus, if a change in the frequency of the wire was detected, it could be interpreted as a change in the strain of the bone, which can help us evaluate the bone healing condition.

The design, concept, and manufacturing of this plate reference from the previous study²⁵.

Experimental setup

The tibia bone was installed on the universal tester, the "Instron Electropuls E10000", to apply axial load. A Synco G1A1 condenser microphone was set up at a perpendicular angle approximately 2 cm away from the center of the sound generator to detect sound. The experimental setup was shown in Fig. 3. Axial load was progressively applied at a rate of 10 N/s, and frequency data was recorded at 0 N, 100 N, 200 N, 300 N, and 400 N of axial load. In each test, the frequency was recorded for 10 times, and the mean frequency of each sample at various axial loadings was calculated. The Audacity program measured the sound frequency, as shown in Fig. 4.

After the IRB approved this study, eight human cadaveric tibia bones donated by patients who had consented before their death to the Department of Anatomy in our hospital were used in the experiment. The tibia was cut 2 cm below the tibial tuberosity and 5 cm above the distal articular surface to use only the diaphysis of tibia. A plate was applied to the midshaft of the cut tibia sample. Six stainless steel cortical screws, diameter 4.5 mm, were

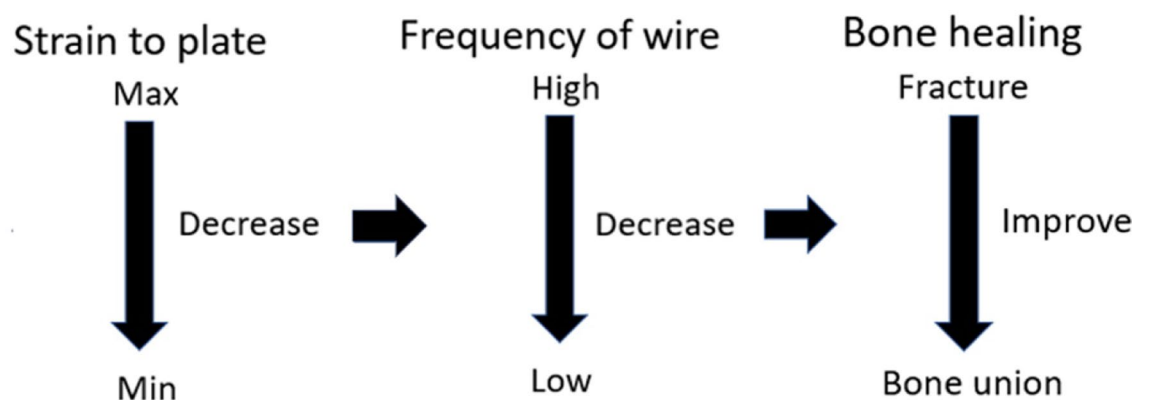


Fig. 1. The concept of the BUDWF plate for detecting bone union through strain change with the changes in the frequency of the wire.

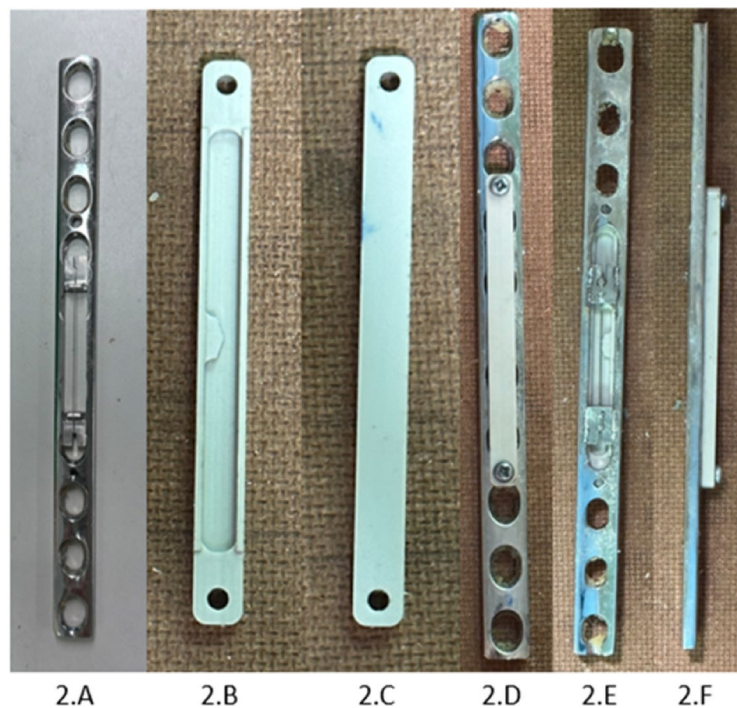


Fig. 2. Images of the BUDWF plate: **2.A**, top view of the plate; **2.B**, **2.C**, bottom and top views of sound generation; **2.D**, **2.E**, and **2.F**, top, bottom, and lateral views of the BUDWF plate.

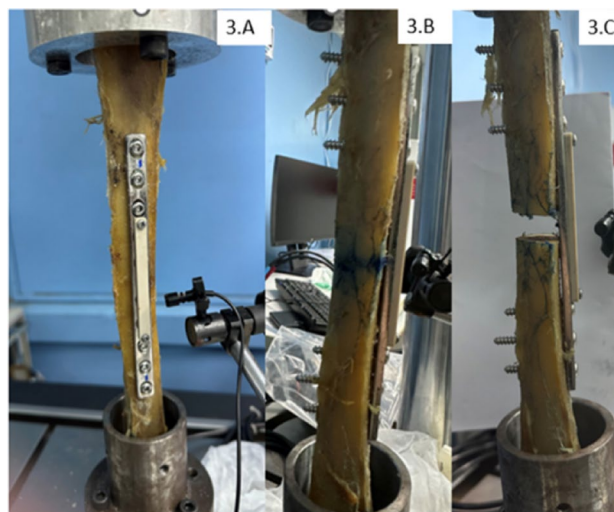


Fig. 3. The experiment setup: **3.A**, Image of tibia bone after fixation with BUDWF plate applied in the “Instron Electropuls E10000”; **3.B**, Test of bone union; **3.C**, Test of bone fracture with a 10 mm gap.

used for fixation, with three screws on each side. The screw was tightened with optimal torque from the outside toward the center on each side. This bone was tested and included in the bone union group. After completing the bone union test, the tibia was cut at the midshaft to create a 10 mm gap at the middle of the wire position. and used for testing in the bone fracture group with a 10 mm gap group. In this study, we conduct experiments comparing only bone healing and bone fracture with a 10 mm gap, and we reference many studies that examine new devices for detecting bone union after fractures^{16–18,21,22,25}.

Frequency data was recorded at 0 N, 100 N, 200 N, 300 N, and 400 N of axial load. Compare the change in frequency at each axial load to that at 0 N in both groups.

Statistical analysis

All data were analyzed using IBM SPSS Statistics version 29. The frequencies are presented with their means and standard deviations. A linear mixed-effects model and a pairwise test with Holm–Bonferroni adjustment for

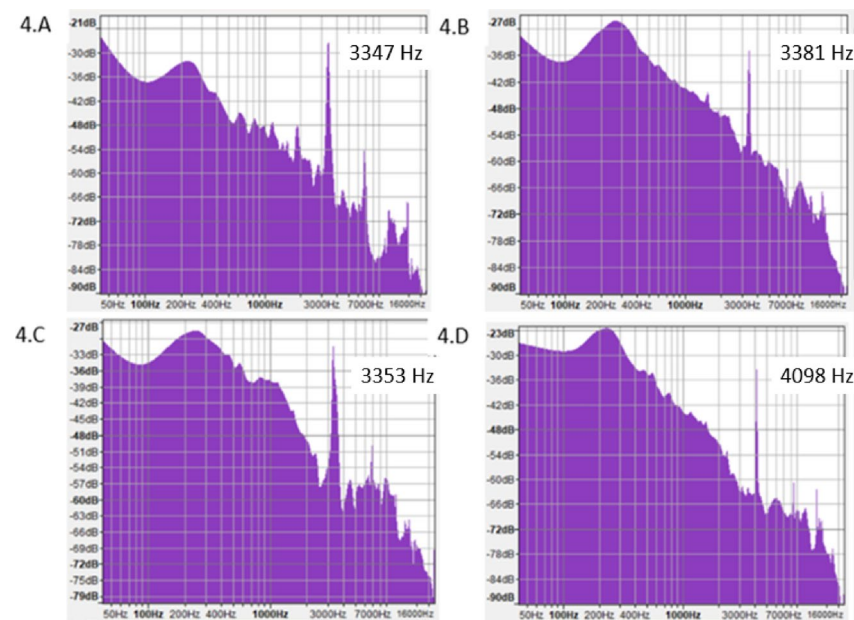


Fig. 4. Frequency analysis of sample No. 1: (4.A) Load 0 N in bone union; (4.B) Load 400 N in the bone union; (4.C) Load 0 N in bone fracture with a 10 mm gap; (4.D) Load 400 N in bone fracture with a 10 mm gap.

Sample No.	Group	Frequency (Hz)					Δ 0 N & 400 N (Hz, %)
		0 N	100 N	200 N	300 N	400 N	
1.	Bone union	3347	3328	3347	3356	3381	34, 1.0%
	Fx with a gap 10 mm	3353	3820	3993	4047	4098	745, 22.2%
2.	Bone union	3322	3322	3332	3366	3381	59, 1.8%
	Fx with a gap 10 mm	3254	3383	3776	4002	4098	844, 25.9%
3.	Bone union	2774	2816	2751	2727	2827	53, 1.9%
	Fx with a gap 10 mm	2759	3042	3466	3712	3781	1022, 37.0%
4.	Bone union	2597	2727	2771	2816	2727	130, 5.0%
	Fx with a gap 10 mm	2582	2510	2945	3413	3781	1199, 46.4%
5.	Bone union	2464	2456	2465	2514	2478	14, 0.6%
	Fx with a gap 10 mm	2456	2409	2514	2909	3245	789, 32.1%
6.	Bone union	2439	2445	2458	2479	2478	39, 1.6%
	Fx with a gap 10 mm	2424	2473	2566	2909	3245	821, 33.9%
7.	Bone union	2395	2375	2474	2474	2434	39, 1.6%
	Fx with a gap 10 mm	2319	2434	2909	3206	3534	1215, 52.4%
8.	Bone union	2357	2343	2349	2434	2434	77, 3.3%
	Fx with a gap 10 mm	2380	2434	2739	3206	3534	1154, 48.5%

Table 1. Mean frequency of wire in each tibial cadaveric sample.

multiple comparisons were used to compare the effects of bone-healing condition and load on Wire Frequency. The ROC Analysis was performed to estimate AUC, Optimal Cut-off Frequency, Sensitivity, and Specificity for Predicting Bone Union at Each Load. The Intraclass Correlation Coefficients and Coefficient of variation (CV) were calculated to evaluate the within-operator repeatability of frequency measurement. A p-value of <0.05 and 95% Confidence intervals was considered statistically significant.

Results

Eight human cadaveric tibial bones, four male and four female, were used in an experiment. The mean age of cadavers was 69.1 years, height was 1.64 m, weight was 64.3 kg, and BMI was 23.8 kg/m². The mean frequency of wire in the bone union group and the bone fracture with a 10 mm gap group (Fx with a gap 10 mm) under various axial loading forces for each sample were shown in Table 1.

The intra-rater reliability of the frequency measurements (Table 2) showed excellent agreement across all loading levels for both the union and fracture conditions. For the bone union group, ICC(3,1) ranged from 0.944

Condition	Load (N)	ICC(3,1)	95% CI	ICC(3,10)	95% CI	Interpretation	CV Mean (%)	CV SD (%)
Union	0	0.944	0.872–0.986	0.994	0.986–0.999	Excellent	3.53	0.88
Union	100	0.950	0.884–0.988	0.995	0.987–0.999	Excellent	3.27	0.73
Union	200	0.959	0.903–0.990	0.996	0.989–0.999	Excellent	2.92	0.37
Union	300	0.953	0.891–0.989	0.995	0.988–0.999	Excellent	3.04	0.48
Union	400	0.949	0.882–0.988	0.995	0.987–0.999	Excellent	3.36	0.53
Fracture	0	0.972	0.932–0.993	0.997	0.993–0.999	Excellent	2.50	0.75
Fracture	100	0.983	0.959–0.996	0.998	0.996–1.000.996.000	Excellent	2.59	0.64
Fracture	200	0.984	0.962–0.996	0.998	0.996–1.000.996.000	Excellent	2.28	0.63
Fracture	300	0.965	0.916–0.991	0.996	0.991–0.999	Excellent	2.37	0.83
Fracture	400	0.946	0.875–0.987	0.994	0.986–0.999	Excellent	2.17	0.40

Table 2. Intraclass correlation coefficients and coefficient of variation (CV) of wire frequency measurements for each condition and Load. ICC(3,1) = Two-way mixed effects, single measurement, absolute agreement. ICC(3,10) = Mean of 10 measurements. CV = (SD/Mean) $\times$ 100.

Parameter	Estimate (Hz)	(95% CI)	<i>p</i> -value
Intercept (Fracture, Load 400 N)	3664.5	(3313.7, 4015.3)	< 0.001
Condition: Union	– 897	(– 938.7, – 855.3)	< 0.001
Load 0	– 973.6	(– 1015.3, – 931.9)	< 0.001
Load 100	– 851.4	(– 893.1, – 809.7)	< 0.001
Load 200	– 551	(– 592.7, – 509.3)	< 0.001
Load 300	– 239	(– 280.7, – 197.3)	< 0.001
Interaction: Union $\times$ Load 0	918	(859.0, 977.0)	< 0.001
Interaction: Union $\times$ Load 100	810.4	(751.4, 869.3)	< 0.001
Interaction: Union $\times$ Load 200	526.9	(467.9, 585.8)	< 0.001
Interaction: Union $\times$ Load 300	242.3	(183.3, 301.2)	0.009

Table 3. Linear Mixed-Effects model: effect of condition and load on wire Frequency. Estimates (Hz) and 95% confidence intervals (CI) from the linear mixed-effects model with specimen as a random effect and condition, load, and their interaction as fixed effects. Positive estimates indicate higher frequency relative to the reference (Fracture at 400 N). Interaction terms show how Union differs from Fracture at each load. * $p < 0.05$ indicates statistical significance.

to 0.959, and ICC(3,10) ranged from 0.994 to 0.996. For the bone fracture group, ICC(3,1) ranged from 0.946 to 0.984, and ICC(3,10) ranged from 0.994 to 0.998. All ICC values were interpreted as “excellent” reliability. Coefficient of variation (CV) analysis demonstrated low measurement variability. In the bone union group, CV means ranged from 2.92% to 3.53%, while the fracture group demonstrated slightly lower variability with CV means between 2.17% and 2.59%. The low CV values across all loads indicate that the repeated measurements produced highly stable frequency readings with minimal dispersion. The wire-frequency measurement method demonstrated excellent repeatability, with ICC(3,1) values > 0.94 and ICC(3,10) values > 0.99 across all loads. CV values remained low (2.17–3.53%), indicating minimal within-operator variability and highly consistent measurements.

The linear mixed-effects model (LMM) (Table 3) demonstrated that wire frequency differed significantly by bone condition and axial load (Condition $\times$ Load interaction, $p < 0.01$). Mean frequency was lower in bone union compared to fractured bones at 400 N by 897 Hz (95% CI: 855.3–938.7 Hz, $p < 0.001$).

Frequency decreased at loads below 400 N in both union and fracture groups. Interaction effects indicated that the reduction in frequency was smaller in the bone union group. Within the bone union group, frequency differences between 0 and 100 N were minimal, whereas frequency increased at 200–300 N ($p < 0.001$). In the bone fracture group, frequency increased progressively from 200 N upward.

The residuals from the linear mixed-effects model were examined to assess the assumption of normality. Both the Kolmogorov–Smirnov test ($D = 0.022$, $n = 800$, $p = 0.200$) and the Shapiro–Wilk test ($W = 0.998$, $n = 800$, $p = 0.563$) indicated no significant deviation from normality. Visual inspection of the Normal Q–Q plot of residuals also showed that the points closely followed the reference line, supporting the assumption of normally distributed residuals. (8 specimen $\times$ 2 condition $\times$ 5 load $\times$ 10 trial = $n = 800$). The graph was shown in Fig. 5.

Pairwise comparisons of estimated marginal means (Table 4) confirmed that wire frequency was significantly higher in the bone fracture group compared to the bone union group at 100 N, 200 N, 300 N, and 400 N (mean differences: –86.6 Hz, –370.1 Hz, –654.8 Hz, and –897.0 Hz, respectively; all $p < 0.001$, Bonferroni-adjusted). No significant difference was observed at 0 N (mean difference: 21.0 Hz, $p = 0.323$). The boxplot graph was shown in Fig. 6. These results indicate that the frequency difference between bone fracture and union becomes more

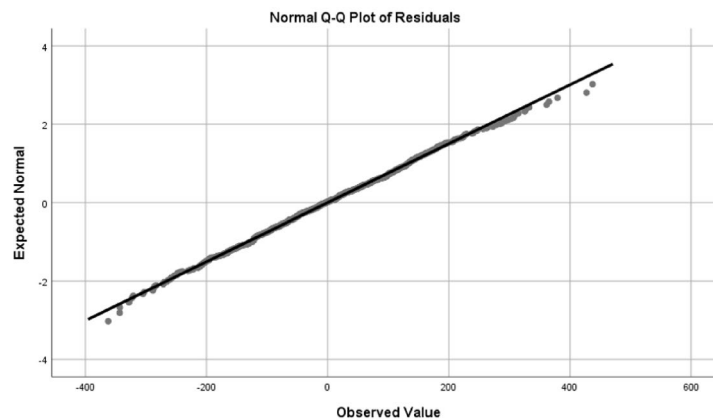


Fig. 5. The model diagnostics.

Force	Fx with a gap 10 mm	Bone union	mean difference	(95% CI)	p-value
0 N	2690.9 ± 406.2	2711.9 ± 406.2	21.00	(− 20.7 , 62.7)	0.323
100 N	2813.1 ± 541.1	2726.5 ± 404.9	− 86.63	(− 128.3 , − 44.9)	<0.001*
200 N	3113.5 ± 561.4	2743.4 ± 396.2	− 370.13	(− 411.8 , − 328.4)	<0.001*
300 N	3425.5 ± 452.1	2770.8 ± 387.9	− 654.75	(− 696.4 , − 613.1)	<0.001*
400 N	3664.5 ± 335.7	2767.5 ± 405.0	− 897.00	(− 938.7 , − 855.3)	<0.001*

Table 4. Comparison of the mean frequency between bone fracture with a 10 mm gap (Fx with a gap 10 mm) and bone union under various axial loading forces. Data are presented as mean ± SD. Pairwise comparisons from the linear mixed-effects model with Bonferroni adjustment are shown. * $p < 0.05$ indicates a significant.

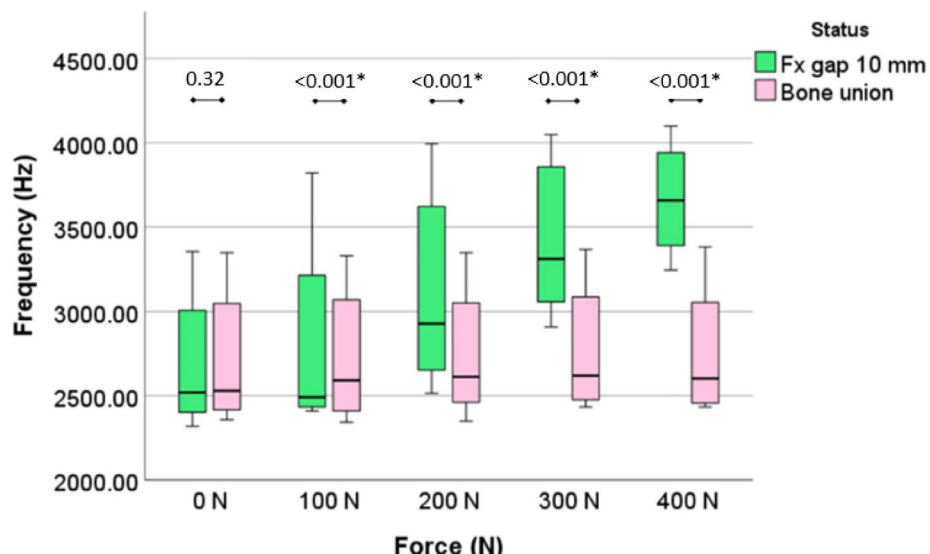


Fig. 6. Boxplot graph comparing the frequency of bone fracture with a gap 10 mm group and bone union group under various axial loading forces. *Significant difference, p -value < 0.05 .

pronounced at higher axial loads. Overall, wire frequency reflects bone healing status most clearly at higher loads.

The pairwise comparison of estimated marginal mean frequencies under different axial loads within each condition (Table 5) showed that the frequency did not significantly change with increasing load from 0 to 400 N in the bone union group (p -values ranging from 0.057 to 1.000), with mean differences ranging from − 58.88 Hz to − 14.63 Hz. In contrast, the Fracture with 10 mm gap group showed a significant and progressive increase in frequency with increasing axial load. Compared with 0 N, frequency increased by 122.25 Hz at 100 N, 422.63 Hz

	Frequency (Hz)	Force	Frequency (Hz)	mean difference	(95% CI)		<i>p</i> -value
Bone union							
Force 0 N	2711.9 ± 406.2	100 N	2726.5 ± 404.9	− 14.625	(− 74.42	, 45.17)	1.000
		200 N	2743.4 ± 396.2	− 31.500	(− 91.29	, 28.29)	1.000
		300 N	2770.8 ± 387.9	− 58.875	(− 118.67	, 0.92)	0.057
		400 N	2767.5 ± 405.0	− 55.625	(− 115.42	, 4.17)	0.090
Fx with a gap 10 mm							
Force 0 N	2690.9 ± 406.2	100 N	2813.1 ± 541.1	− 122.25	(− 182.04	, − 62.46)	<0.001*
		200 N	3113.5 ± 561.4	− 422.63	(− 482.42	, − 362.83)	<0.001*
		300 N	3425.5 ± 452.1	− 734.63	(− 794.42	, − 674.83)	<0.001*
		400 N	3664.5 ± 335.7	− 973.63	(− 1033.42	, − 913.83)	<0.001*

Table 5. Comparison of mean frequency under axial loading of 100 N, 200 N, 300 N, 400 N, and 0 N in each group. Estimated marginal means from a linear mixed-effects model with specimen as a random effect and load as a fixed effect. Pairwise comparisons were Bonferroni-adjusted. * $p < 0.05$ indicates statistical significance.

Force	Fx with a gap 10 mm Δ frequency (Hz, %)	Bone union Δ frequency (Hz, %)	p-value	Holm–Bonferroni adjusted p-value
0 N & 100 N	122.3, 4.5%	14.6, 0.5%	0.18	0.18
0 N & 200 N	422.6, 15.7%	31.5, 1.2%	0.003*	0.006*
0 N & 300 N	734.6, 27.3%	58.9, 2.2%	< 0.001*	0.003*
0 N & 400 N	973.6, 36.2%	55.6, 2.1%	< 0.001*	0.004*

Table 6. Comparison of mean frequency under various axial loading forces and frequency change when comparing 0 N in bone fracture with a 10 mm gap (Fx with a gap 10 mm) and bone union. *Significant difference, p -value < 0.05.

Load (N)	AUC (95% CI)	p-value	Cut-off Frequency (Hz)	Sensitivity (95% CI)	Specificity (95% CI)	PPV (95% CI)	NPV (95% CI)	Accuracy (95% CI)
All Loads Combined	0.697 (0.580–0.814)	0.002	≤ 2868	75.0% (58.80–87.31.80.31)	65.0% (48.32–79.37)	68.18% (57.53–77.22)	72.22% (59.21–82.32)	70.0% (58.72–79.74)
0	0.453 (0.161–0.745)	0.753	–	–	–	–	–	–
100	0.563 (0.268–0.857)	0.674	–	–	–	–	–	–
200	0.750 (0.504–0.996)	0.093	–	–	–	–	–	–
300	0.875 (0.694–1.000.694.000)	0.012	≤ 2863	75.0% (34.91–96.81)	100% (63.06–100.0)	100% (54.07–100.0)	80.0% (54.64–93.0.64.0)	87.5% (61.65–98.45)
400	0.938 (0.822–1.000.822.000)	0.003	≤ 3036	75.0% (34.91–96.81)	100% (63.06–100.0)	100% (54.07–100.0)	80.0% (54.64–93.0.64.0)	87.5% (61.65–98.45)

Table 7. ROC analysis results: AUC, optimal Cut-off Frequency, Sensitivity, and specificity for predicting bone union at each Load.

at 200 N, 734.63 Hz at 300 N, and 973.63 Hz at 400 N (all $p < 0.001$, Bonferroni-adjusted). These results indicated that axial loading substantially increased vibrational frequency in fractured bones, whereas frequency remained relatively stable in bone union.

Because the comparison is within the same condition (Fx with gap 10 mm or Union) and the purpose is to summarize the Δ frequency at each load, a simple pairwise test with Holm–Bonferroni adjustment for multiple comparisons is sufficient and statistically appropriate. Using LMM for these comparisons would not change the interpretation, and including them would be redundant, as the primary inferential analysis for between-condition differences is already presented in Tables 4 and 5 using LMM.

Therefore, we report Table 6 with Holm–Bonferroni-adjusted p -values to control for multiple comparisons. This approach preserves clarity and directly addresses the research question of frequency change under increasing load.

ROC Analysis for Predicting Bone Union was shown in Table 7; Fig. 7. Wire frequency could predict bone union when all loads were combined (AUC 0.697, 95% CI: 0.580–0.814, $p = 0.002$) with an optimal cut-off ≤ 2868 Hz (sensitivity 75.0%, specificity 65.0%, accuracy 70.0%). When analyzed by individual loads, wire

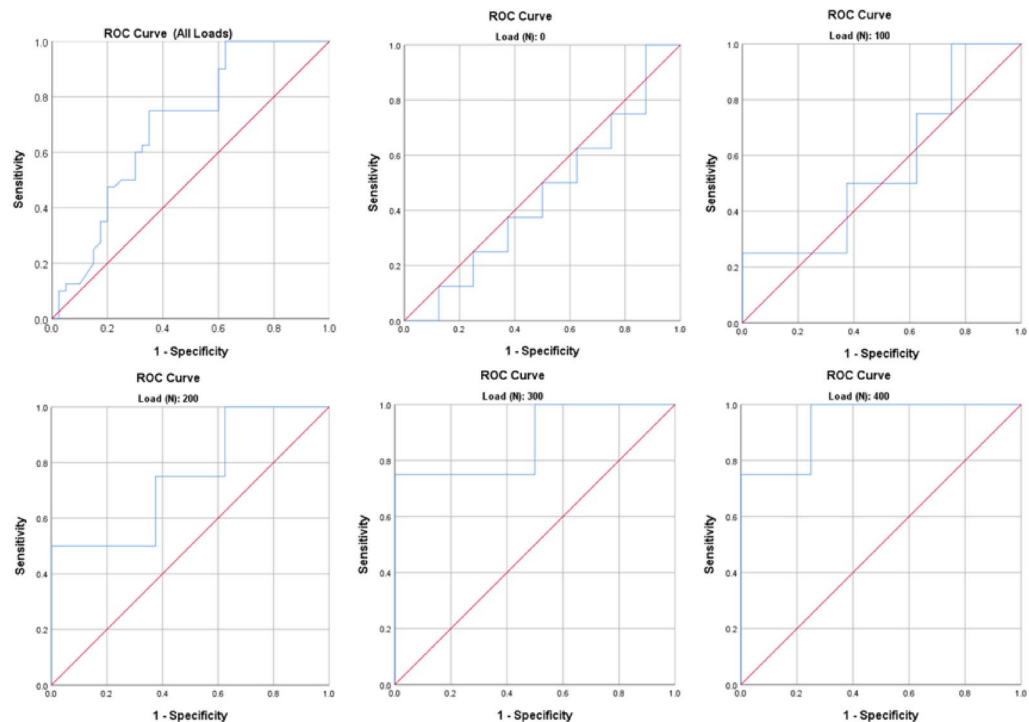


Fig. 7. The ROC Analysis.

frequency did not reliably predict union at 0, 100, or 200 N (non-significant AUCs). Prediction was accurate at 300 N (AUC 0.875, cut-off ≤ 2863 Hz) and 400 N (AUC 0.938, cut-off ≤ 3036 Hz), with sensitivity 75.0%, specificity 100%, and accuracy 87.5%. In summary, wire frequency is most predictive of bone union under higher axial loads (≥ 300 N).

Discussion

This study demonstrated that the frequency change of wire on the BUDWF plate can assess bone healing. The frequency change under axial loads of 400 N and 0 N was significantly different when comparing the bone fracture group, which had a 10 mm gap, with the bone union group. Although these results were supported by the findings of a previous study that tested the BUDWF plate in sawbone models²⁵, this study strongly confirmed the efficacy of the BUDWF plate in detecting bone union in cadaveric bones, specifically through experiments on cadaveric tibial bones using a larger sample size.

Although previous studies showed various methods and devices invented to combine plates for assessing bone healing, such as strain gauges^{16,17}, displacement of tungsten¹⁸, fluid level¹⁹ or contrast media displacement²⁰, PZT voltage²¹, wavelengths²², resonance response frequency (RRF)²³, force sensor²⁴, however, the main concept of these methods was to detect changes in strain on the plate, similar to the BUDWF plate, with the difference being the technique used for strain detection; this plate utilized frequency change for detection.

The bone union detection using a wire's natural frequency (BUDWF) plate in this study identifies changes in the strain of the implant to assess bone healing. When a bone fracture occurs, immediate fixation with the plate and screw is applied under axial load, primarily transferring the load through the plate at the fracture site, which increases the strain on the wire in the middle of the plate and causes the wire's frequency to be high. As the bone gradually heals and is subjected to axial load again, the load begins to be distributed between the bone and the plate, leading to a decrease in the strain on the plate and the frequency of the wire. Ultimately, when the bone is fully healed, the axial load will mainly be supported by the bone rather than the plate, indicating minimal strain on the plate and frequency of the wire.

According to the bone union group results, the frequency remained consistent under various axial loads because the strain on the plate did not change. In contrast, there was a change in frequency in the bone fracture with a 10 mm gap group when subjected to different axial loads. As shown in Table 4; Fig. 6, this confirms that frequency measurements can detect changes in strain and assess bone healing. The magnitude of axial load significantly affects the frequency change in bone fractures with a 10 mm gap, as higher axial loads can increase the strain on the plate more than lower axial loads. As shown in Table 6.

The mean frequency of both groups under an axial load of 0 N (unloaded condition) and the mean frequency of various axial loads in the bone union group are not significantly different. This result confirms that the frequency of the wire after an immediate bone fracture treated with a plate and screw in the unloaded condition should be similar to that observed at bone union. This indicates that a gradual decrease in wire frequency near the unloaded condition suggests bone union.

The results showed that the high axial load was significantly different between the two groups, especially for loads greater than 200 N, 300 N, and 400 N. The load-sharing transition can explain these results: Most of the axial load was absorbed by the bone, leaving the remaining force acting on the plate, which led to changes in the wire's strain and tension. Therefore, in the bone union condition, there was no significant difference in frequency change between low and high axial load because most of the force was absorbed by the bone, and the stiff had a low force impact on the plate. Meanwhile, in a bone fracture, most of the force acts directly on the plate, with high loads creating high strain and tension in the wire, resulting in a significant frequency change. However, they had many factors that affect the force on the strain and tension, such as tibia geometry, which can increase or decrease the tension of the wire depending on the curvature of the cortex surface and the tightness of the screw, the plate's stiffness, which influences how the strain and tension change under the same force. Higher stiffness results in less Deformation, and minimal changes in strain and tension lead to a smaller change in frequency. The wire tension-strain nonlinearity, which can lead to wire failure at high strains, was considered in the design of the BUDWF plate. It was designed so that the wire remains within its elastic limit, meaning elastic deformation, and the tension-strain nonlinearity in the tension-strain curve does not occur.

Therefore, this study recommends detecting bone healing by monitoring the decrease in frequency under an axial load of 400 N, which shows a significant difference compared to the other axial loads and approaches the unloaded condition (axial load of 0 N). This can help determine the healing of the bone union.

Limitations and future works

This study had several limitations. First, this study was a cadaveric study that examined only one fracture pattern, which does not accurately reflect the normal bone healing process. Many factors influence healing, such as the fracture pattern, type of implant fixation, and number of screws, which may affect outcomes related to axial load, tension, strain, and frequency. Further research is needed in the future. Second, it was a cadaveric study using tibial cadaveric bones without soft tissue coverage that could interfere with the wire's vibration. Nonetheless, this study can demonstrate the efficacy of the BUDWF plate in assessing bone healing, and further examinations should be conducted in the future. Third, the initial frequency of the wire on the BUDWF plate varied in each sample due to the laser welding process. However, it still remained within the planned frequency range of 2000 to 3500 Hz, and our results indicated that all of the BUDWF plates continued to function in detecting frequency changes. Finally, this study uses only manual excitation without a controlled actuator to generate sound, which may not be standard. The force of manual pressure may vary, but based on the concept of natural frequency—which does not change depending on the amplitude of sound—different pressure forces do not affect the frequency change like amplitude does, as confirmed by the excellent agreement of the intraclass correlation coefficients.

The benefit of the BUDWF plate was its safety for the patient. Made from medical-grade stainless steel 316 L, it showed biocompatibility with the body and did not need a circuit or battery like the others. However, this plate is still the prototype, which needs to develop in many areas, such as the size of the implant that could irritate the soft tissue or skin, the complexity of manufacturing, and the lack of data about fatigue failure, the effect of soft tissue around the fracture, temperature and sterilization, and the difficulty in detecting sound within the body and conduct experiments with various patterns of bone healing, not just fractures with a 10 mm gap, to confirm the efficacy of this plate.

We hope this design development makes the plate easier to use and safer.

Conclusions

Although the BUDWF plate design is straightforward and does not depend on advanced technology, it can evaluate bone healing through frequency measurement. The authors suggest monitoring bone healing by tracking the decrease in frequency under an axial load of 400 N as it approaches the unloaded condition (axial load of 0 N). This can assist in determining the healing of the bone union. The authors hope this study marks the beginning of developing an implant that can treat bone fractures and detect bone union by itself.

Data availability

The data supporting this study's findings are available from the corresponding author (P.T.) upon reasonable request.

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Author contributions

Chavarin Amarase, Chanyaphan Virulsri, and Pairat Tangpornprasert contributed to the conception and design of the study. Chavarin Amarase and Pisitpong Chancharoen prepared the materials and collected and analyzed the data. Chavarin Amarase wrote the first draft and edited the manuscript. All the authors reviewed and approved the final manuscript.

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Declarations

Competing interests

The authors declare no competing interests. Each author certifies that there are no commercial associations (consultancies, stock ownership, equity interest, patent/licensing arrangements, etc.) that might pose a conflict of interest in connection with the submitted article related to the author or any immediate family members.

Ethical approval

This study was approved by the Ethical Committee of the Faculty of Medicine, Chulalongkorn University (No 0395/65).

Additional information

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